

## M002 Week 3 — AND Conditions

**Topic:** AND — two things both true · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

**AND** checks two things. The action runs only when **both** are true.

### 1 · Predict

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
    agent.turn(LEFT_TURN)
```

Both true. What happens?

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```
if not agent.detect(BLOCK, FORWARD) and not agent.detect(BLOCK, DOWN):
    agent.move(FORWARD, 1)
```

Path ahead is clear, but a block is below. Does it move? Why?

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## 2 · Trace

```
while True:
    if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
        agent.turn(LEFT_TURN)
    else:
        agent.move(FORWARD, 1)
```

At a corner (block ahead AND block right), what runs?

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At an open path, what runs?

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### 3 · Spot and Fix Bugs 🐛

**Bug A** — Turn left only when block ahead AND block on right.

```
if agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Fix:

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Why wrong?

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**Bug B** — Both checks should be different: front and right.

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Fix:

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Why wrong?

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#### 4 · Write Code

Move forward only when the path ahead is clear AND the path on the left is clear. Write it.

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## 5 · Finish the Maze 📷

Solve your homework maze using one or more **AND** conditions. Send a photo or video of the agent at the end. Then finish these.

First, I ...

I used **AND** when ...

Both were true when ...

One tricky moment was ...

To fix it, I ...

## 6 · Record Your Walkthrough 🎥

Plan your phone video, then record. Try these words: **AND**, **condition**, **both**, **true**, **turn**.

1. Show the start. Run your code.
2. Read each `if ... and ...` block out loud. Say what makes it run.
3. Show one spot where the agent did nothing because only one check was true.
4. Say what **AND** means.

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**Submit** ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.